VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS201

Data Structures and Algorithmic Analysis - Semester Exam Question Bank &

Syllabus Focus: CS201 High-Yield Semester Exam Question Bank & Model Solutions

Summary: Comprehensive question bank with previous semester university examination questions, m

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Linear Structures & Computational Complexity

" Syllabus Scope: Asymptotic notation O, Omega, Theta, dynamic arrays, singly and doubly linked lists, am

Unit 2: Stacks, Queues & Recursion Mechanics

" Syllabus Scope: Stack evaluation, infix-to-postfix conversion, circular queues, double-ended queues deq

CS201: Data Structures and Algorithmic Analysis - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Hierarchical Trees & Balanced Search Trees

" Binary Search Trees BST, AVL rotations, Red-Black tree properties, B-Trees and B+ Trees node splitting

Unit 4: Priority Queues, Heaps & Hashing

" Binary min/max heaps, heap sort, hash functions, open addressing linear, quadratic, chaining with linked lis

Unit 5: Graph Traversal & Greedy Networks

" Adjacency list/matrix, Breadth-First Search BFS, Depth-First Search DFS, Dijkstra shortest path, Prim and Kr

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Prove the time complexity of QuickSort average-case On log n versus worst-case On^2, explaining pivot
2. Construct an AVL tree by sequentially inserting elements [15, 20, 24, 10, 13, 7, 30, 36, 25], illustra
3. Apply Dijkstra Single Source Shortest Path algorithm on a directed weighted graph with 6 vertices, sho

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Thomas H. Cormen, Charles E. Leiserson - Introduction to Algorithms MIT Press 3rd Ed
* Mark Allen Weiss - Data Structures and Algorithm Analysis in C++ Pearson 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students